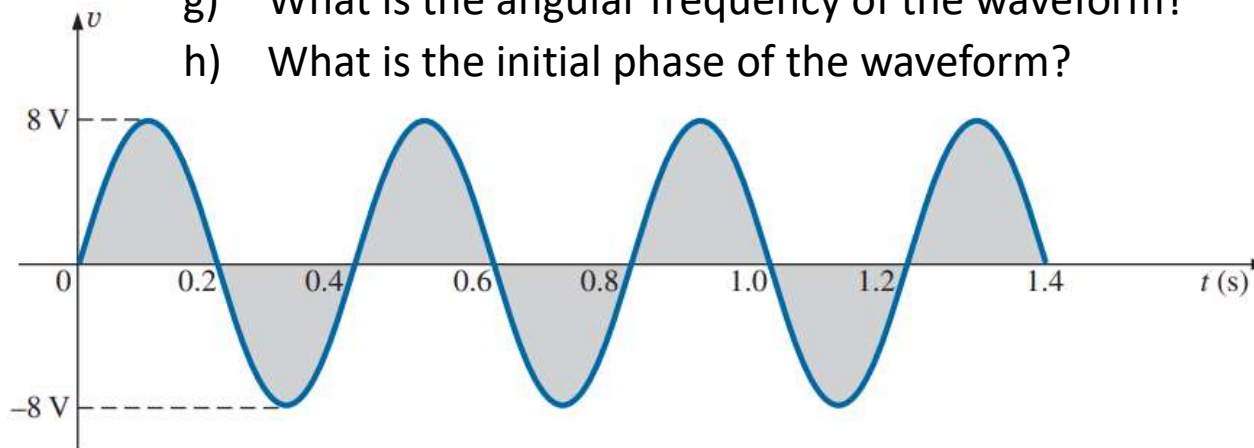


Problem 2

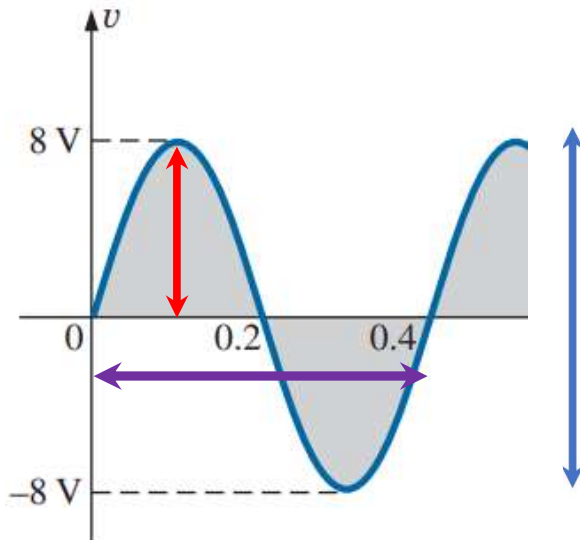
- For the sinusoidal waveform shown below,
 - What is the peak value?
 - What is the instantaneous value at 0.3s and 0.6s?
 - What is the peak-to-peak value of the waveform?
 - What is the period of the waveform?
 - How many cycles are shown?
 - What is the natural frequency of the waveform?
 - What is the angular frequency of the waveform?
 - What is the initial phase of the waveform?



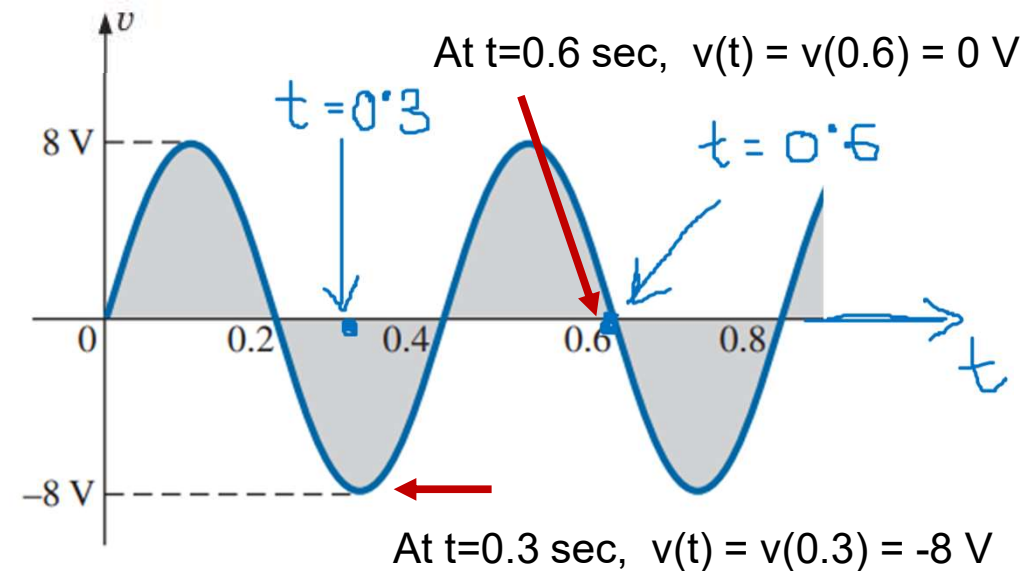
Ans: Try yourself

(a) What is the peak value (amplitude) ? $A_m = 8 \text{ V}$

(c) What is the peak-to-peak value? $V_{p-p} = 16 \text{ V}$



(b) What is the instantaneous value at 0.3s and 0.6s ?



(d) (Time) Period of the waveform? $T = 0.4 \text{ sec}$
 [time it takes to complete one full cycle]

(h) Initial Phase of the waveform? $\phi = 0^\circ$ (0 degree)

[as the Curve (sin) is starting from origin as it's supposed to be]

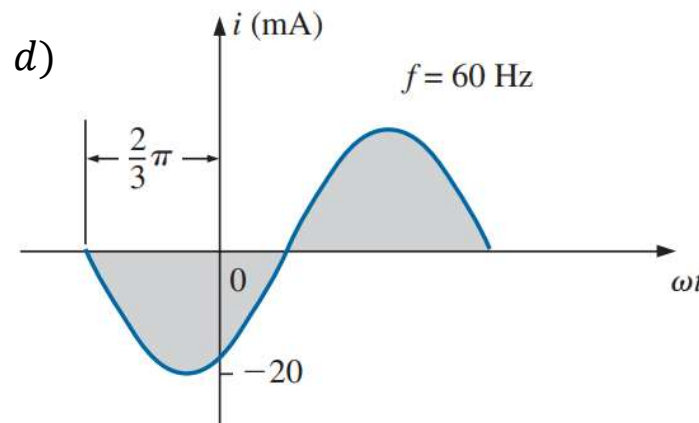
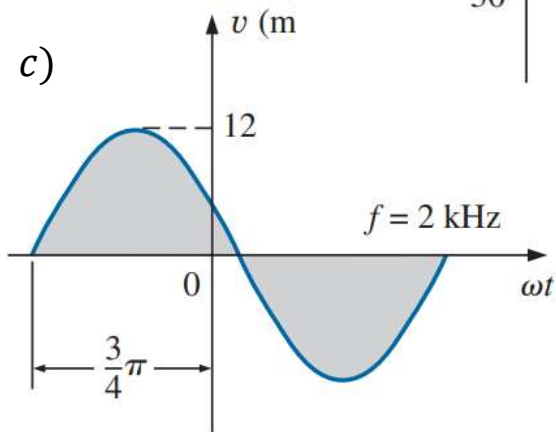
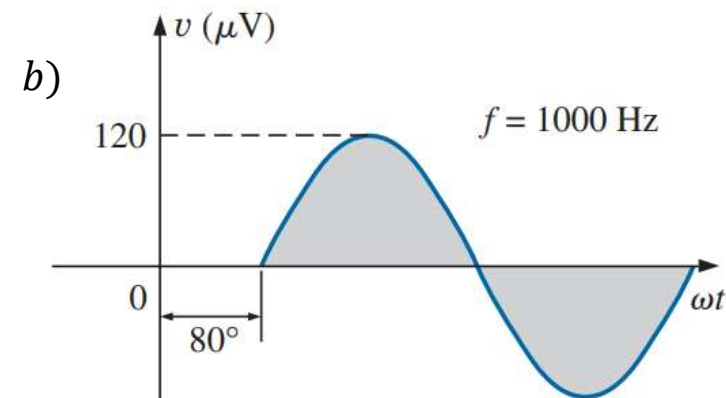
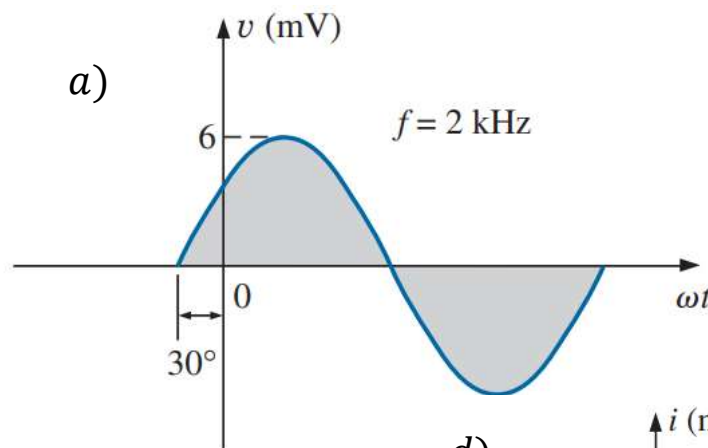
(e) How many cycles? = **3.5 full cycles or 7 half cycles**

(f) Natural frequency? $f = 1 / T$
 $= (1 / 0.4) = 2.5 \text{ Hz}$

(g) Angular frequency? $\omega = 2 \cdot \pi \cdot f$
 $= 5 \pi \text{ rad/s}$

Problem 4

- Write analytical expressions for the waveforms with the initial phase in degrees.



Ans:

a) $6 \times 10^{-3} \sin(2\pi 2000t + 30^\circ) \text{ V}$

b) $120 \times 10^{-6} \sin(2\pi 1000t - 80^\circ) \text{ V}$

c) $12 \times 10^{-3} \sin(2\pi 2000t + 135^\circ) \text{ V}$

d) $20 \times 10^{-3} \sin(2\pi 60t - 60^\circ) \text{ A}$

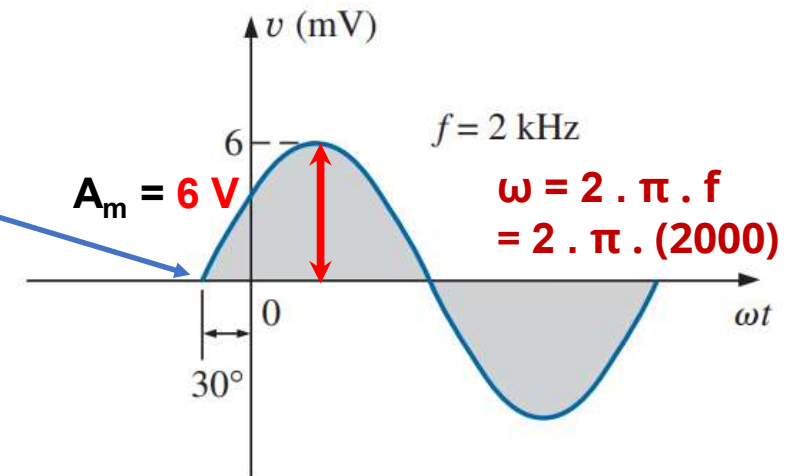
Let's assume, our general wave equation to be : $A_m \sin(\omega t + \varphi)$

(a)

as the sin curve is starting from 30° left of origin (positive),
But usually sin curves are supposed to start from origin line.

$$\text{So, } \varphi = 30^\circ$$

$$A_m \sin(\omega t + \varphi) = 6 \sin(2\pi \cdot 2000 t + 30^\circ)$$



(d)

as we're thinking about sin curve, let's think from here that the curve has started (start of positive half cycle of sin curve). We see that it's right of the origin. So, φ will be negative. Let's calculate that -

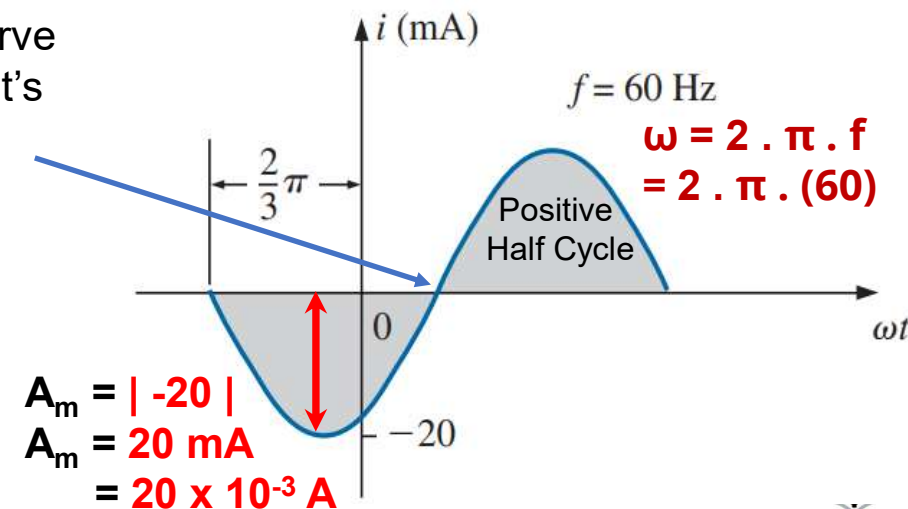
Whole negative half cycle is suppose to be 180° or (π) .

$$\text{So, } \varphi = 180 - 120 = 60^\circ \text{ or, } \varphi = (\pi - 2\pi/3)$$

$$\text{So, } \varphi = \pi/3$$

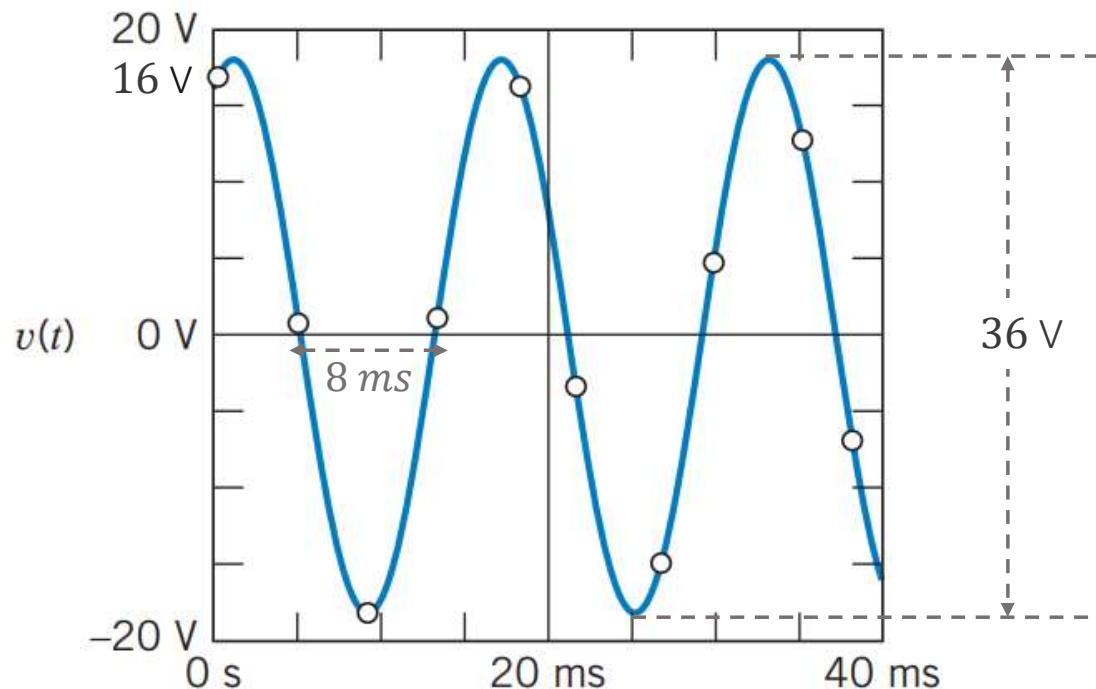
$$\text{Thus, } \varphi = -\pi/3 \text{ or, } -60^\circ$$

$$A_m \sin(\omega t + \varphi) = 20 \sin(2\pi \cdot 60 t - \pi/3)$$

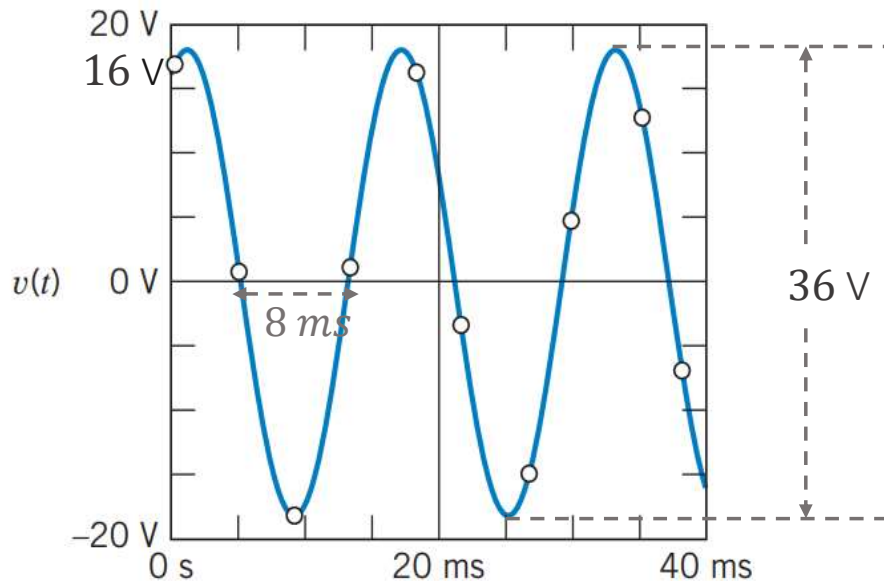


Problem 5

- The following figure shows a sinusoidal voltage $v(t)$, plotted as a function of time t . Represent $v(t)$ by a function of the form $A_m \cos(\omega t + \varphi)$ with φ in degrees.



Ans: $v(t) = 18 \cos(393t - 27^\circ) \text{ V}$



$$A_m \cos(\omega t + \varphi)$$

Here, peak-to-peak voltage is given = $V_{p-p} = 36 \text{ V}$

We know, Amplitude is Half of V_{p-p} , so $A_m = 36 / 2 = 18 \text{ V}$

Here, we can see one negative half cycle (given in the figure) is 8 ms

So, full cycle would be $T = 2 \times 8 \text{ ms} = 16 \text{ ms} = 16 \times 10^{-3} \text{ s}$

Thus natural frequency, $f = 1 / T = 1 / 16 \times 10^{-3} = 62.5 \text{ Hz}$

Thus, angular frequency, $\omega = 2 \cdot \pi \cdot f = 2 \cdot \pi \cdot (62.5) = 393 \text{ rad/s}$

Thus, till now we have, $v(t) = 18 \cos(393 \cdot t + \varphi)$
[we yet need to find φ]

At $t = 0 \text{ s}$, we see our $V(t=0)$ shows initial value of 16 V

Thus putting $t=0 \text{ s}$, $16 = 18 \cos(393 \cdot 0 + \varphi)$

$$16/18 = \cos \varphi$$

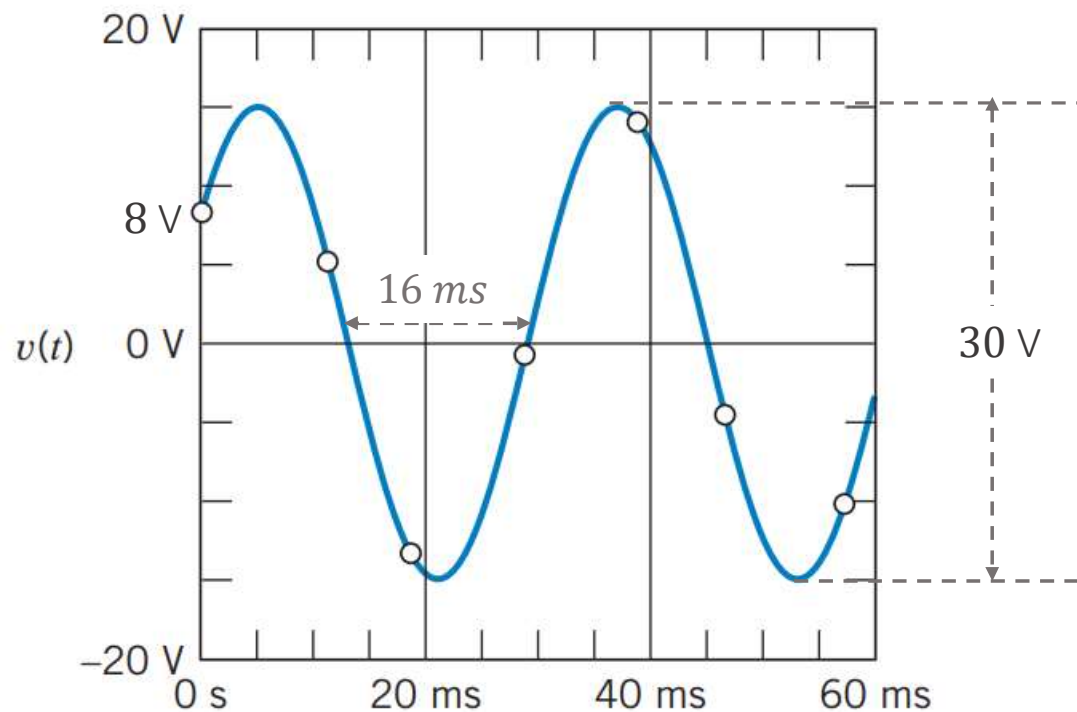
$$\varphi = \cos^{-1}(16/18)$$

$$\varphi = 27^\circ$$

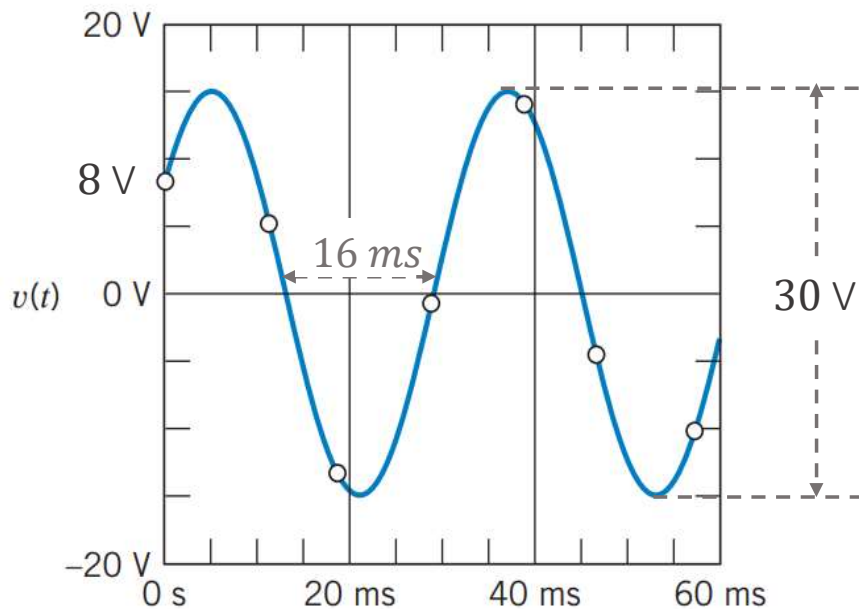
Thus, answer = $18 \cos(393t - 27^\circ)$

Problem 6

- The following figure shows a sinusoidal voltage $v(t)$, plotted as a function of time t . Represent $v(t)$ by a function of the form $A_m \sin(\omega t + \varphi)$ with φ in degrees.



Ans: $v(t) = 15 \sin(196t + 32^\circ) \text{ V}$



$$A_m \sin(\omega t + \varphi)$$

Here, peak-to-peak voltage is given = $V_{p-p} = 30 \text{ V}$

We know, Amplitude is Half of V_{p-p} , so $A_m = 30 / 2 = 15 \text{ V}$

Here, we can see one negative half cycle (given in the figure) is 8 ms

So, full cycle would be $T = 2 \times 16 \text{ ms} = 32 \text{ ms} = 32 \times 10^{-3} \text{ s}$

Thus natural frequency, $f = 1 / T = 1 / (32 \times 10^{-3}) = 31.25 \text{ Hz}$

Thus, angular frequency, $\omega = 2 \cdot \pi \cdot f = 2 \cdot \pi \cdot (31.25) = 196 \text{ rad/s}$

Thus, till now we have, $v(t) = 15 \sin (196 \cdot t + \varphi)$
[we yet need to find φ]

At $t = 0 \text{ s}$, we see our $V(t=0)$ shows initial value of 8 V

Thus putting $t=0 \text{ s}$, $8 = 15 \cos (196 \cdot 0 + \varphi)$

$$8/15 = \sin \varphi$$

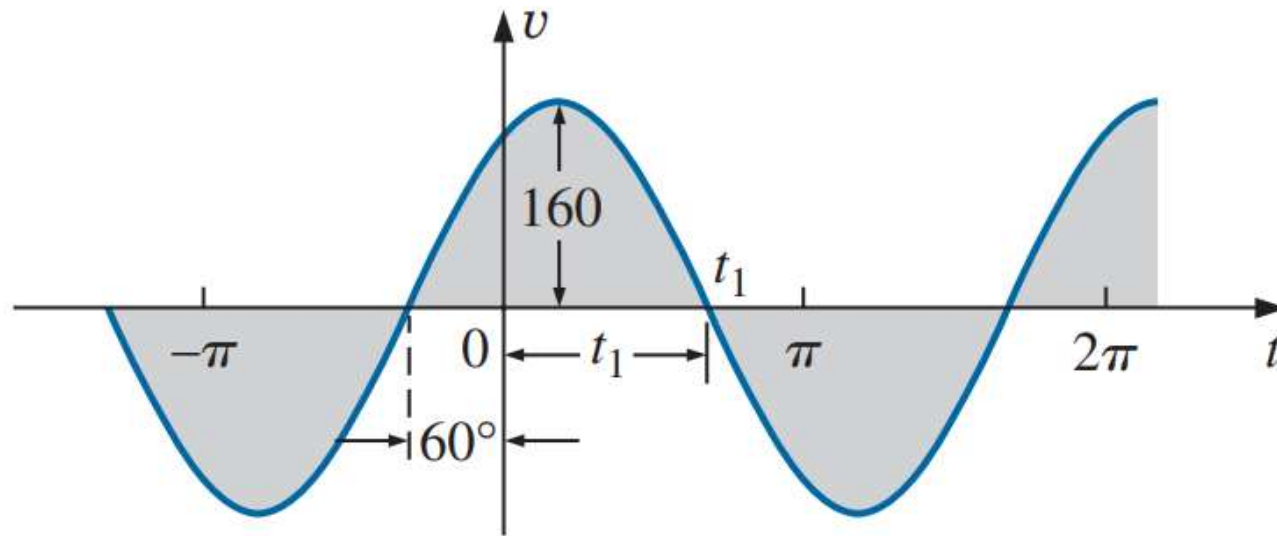
$$\varphi = \sin^{-1} (8/15)$$

$$\varphi = 32.23^\circ$$

Thus, answer = $15 \sin (196t + 32^\circ)$

Problem 7

- A sinusoidal voltage $v(t) = 160 \sin(2\pi 1000t + 60^\circ)$ is plotted as a function of time t below. Determine the time t_1 when the waveform crosses the axis.



Ans: $t_1 = 0.33 \text{ ms}$

$$v(t) = 160 \sin (2\pi 1000t + 60^\circ)$$

$$0 = 160 \sin(2\pi 1000t_1 + 60^\circ)$$

$$0/160 = \sin(2\pi 1000t_1 + 60^\circ)$$

$$0 = \sin(2\pi 1000t_1 + 60^\circ)$$

$$(2\pi 1000t_1 + 60^\circ) = \sin^{-1} 0 = \pi$$

[from figure we can see that, **the end of Positive half cycle is at t_1** which generally occurs for sin curves at π and $(\sin \pi) = 0$]

$$2\pi 1000t_1 = \pi - \pi/3 \quad [60^\circ = \pi/3]$$

$$t_1 = (2\pi/3) \div (2\pi 1000)$$

$$t_1 = 3.33 \times 10^{-4} \text{ sec}$$

$$t_1 = 0.33 \text{ ms}$$

